

smol-kimi-k3

Repository

Role fit: Research, AI Research, Machine Learning, ML Systems

Skills: Python, PyTorch, Triton, CUDA, Kimi Delta Attention, Gated MLA, MoE, Muon

Training-oriented 49M-parameter reproduction of the Kimi K3 language backbone that runs on a single 8GB GPU.

- Reproduced a 49,019,770-parameter Kimi K3-inspired language model with 9 Kimi Delta Attention layers, 4 Gated MLA layers, Attention Residuals, and Stable LatentMoE.
- Trained the model from scratch on TinyStories using BF16 on a single RTX 4060 Laptop GPU; selected the step-9,750 checkpoint at 1.6277 validation loss.
- Integrated chunkwise Triton KDA kernels, cached KDA/MLA autoregressive decoding, Muon and Per-Head Muon optimization, and an FP32 recurrent CPU fallback.
- Built atomic checkpointing and a live training dashboard for loss, throughput, VRAM, and samples; verified architecture, causality, decoding, and artifacts with 6 tests.